

Prop convergencia uniforme borde imp convergencia uniforme interior

Proposición 1. Sea $\Omega \subset \mathbb{C}$ un dominio acotado y sean $(f_n)_{n \in \mathbb{N}} \subset \mathcal{H}(\Omega) \cap \mathcal{C}(\overline{\Omega})$ y $f \in \mathcal{H}(\Omega) \cap \mathcal{C}(\overline{\Omega})$. Entonces, $f_n \xrightarrow[n \rightarrow \infty]{\partial\Omega} f \implies f_n \xrightarrow[n \rightarrow \infty]{\overline{\Omega}} f$.

Demostración: Sea $\varepsilon > 0$, por hipótesis, $\exists N \in \mathbb{N} : \forall n \geq N : \max_{z \in \partial\Omega} |f_n(z) - f(z)| < \varepsilon$. Como $f_n - f$ es holomorfa en Ω y continua en $\overline{\Omega}$, por el , $\forall n \geq N : \max_{z \in \overline{\Omega}} |f_n(z) - f(z)| = \max_{z \in \partial\Omega} |f_n(z) - f(z)| < \varepsilon$. Por tanto, $f_n \xrightarrow[n \rightarrow \infty]{\overline{\Omega}} f$. ■

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